TECHNICAL REPORT · FTALGO 0.2

Factoring and chemistry on a simulated fault-tolerant quantum computer

Algorithms run end to end on a surface-code machine whose logical error rates were measured by circuit-level simulation

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the explainer site · github.com/Jaspersands/ft-algorithms ·

SUMMARY

Shor's algorithm (full modular exponentiation in both textbook Cuccaro and modern windowed measurement-based-uncomputation arithmetic, semiclassical approximate QFT) and iterative phase estimation of molecular ground-state energies (H_2, HeH^+, and frozen-core LiH across its dissociation coordinate) are run end to end on a simulated rotated-surface-code machine. Every logical error channel, idling, lattice-surgery merges and the surgery CNOT, was measured in 316 circuit-level experiments (0.82×10^9 shots) with stabilizer-qec under SD6 noise and extracted with baselines propagated through each operation; magic states come from published factory models. Whole surgery experiments predicted from their parts agree with circuit-level measurements to 0.93–1.26× (median 1.04×). Modern arithmetic cuts Toffolis by up to 2.7–4.0×, reducing rounds by 1.4–1.8× on the simulator and dropping the distance needed for 90% peak probability on N = 15 from d = 13 to d = 11. Biased-noise compilation under asymmetric XZZX surface codes (d_X × d_Z) reduces the physical qubit footprint by 3.4× at dephasing bias η = 100 and 4.25× at η = 500 for RSA-2048. Extrapolated with the same error model to RSA-2048, textbook arithmetic requires d = 35, 32.5 M physical qubits, and 1.96 years; modern windowed arithmetic lowers run time to 298 days (43.2 M physical qubits). For Gidney's 2025 algorithm, an empirical sensitivity analysis demonstrates that our measured circuit-level noise requires d = 29 (vs. his assumed 25) because SD6 noise decodes to an idle logical error rate 4.3× higher at d = 25 than Gidney's uncalibrated assumption.

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1 Introduction

Resource estimates for fault-tolerant algorithms usually multiply a gate count by an assumed logical error rate. This work runs the algorithms instead. A logical-level simulator (ftsim, written for this project) executes the actual circuits, with every operation followed by its

measured logical channel and every waiting patch accumulating measured idle error. The results are success probabilities as functions of code distance, physical error rate and magic-state source, with error budgets that say which operations cause the failures.

The approach rests on one fact and one approximation. The fact: under Pauli circuit noise, for a Clifford circuit decoded into a Pauli-frame correction, the logical action of one surface-code operation is exactly a Pauli channel, since every syndrome history ends in some logical Pauli. The approximation: consecutive operations' channels are composed as independent. Section 4.3 measures how good that is.

Contributions:

- A calibrated logical model of a rotated surface code under SD6 noise: idle (including logical Y), Z⊗Z and X⊗X merges and the surgery CNOT, from 316 experiments, with fits in d and p.
- A composition test: whole lattice-surgery experiments predicted from parts at the logical level and compared with circuit-level measurement.
- Shor's algorithm with honest arithmetic (no use of the period anywhere in the circuit) run on that model for N up to 143, scored by peak probability because, for small N, uniformly random outcomes already "factor" with probability 1.00 (N = 15).
- Modern fault-tolerant arithmetic implemented and executed on the simulator: Gidney (2018) carry uncomputation via mid-circuit measurement and classical feed-forward CZ fixup, algebraic normal form table lookups, and windowed modular exponentiation, cutting Toffolis by 2.7–4.0×.
- Molecular phase estimation from first-principles Hamiltonians to chemical accuracy, scaling from minimal diatomics (H_2, HeH^+) to Lithium Hydride (LiH) with active-space frozen core, Z_2 spin-parity tapering, and full dissociation curve tracking.
- Biased noise compilation under asymmetric rectangular XZZX surface codes (d_X × d_Z), cutting physical qubit footprints by 3.4× to 4.25× at dephasing bias $\eta \in [100, 500]$.
- Interactive 2D lattice surgery fast-block floorplan visualizer and WebAssembly client-side QPE quantum simulator.
- Robustness and sensitivity analysis of the RSA-2048 code distance ($d = 29$ vs. $d = 25$), isolating why circuit-level SD6 noise requires $d = 29$ under calibrated physical simulation.

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2 Methods

2.1 Circuit-level calibration

All physical-level simulation uses stabilizer-qec 1.2 (the author's engine, which matches Stim and PyMatching) with SD6 noise at $p \in \{0.1, 0.2, 0.3, 0.5\}\%$ and correlated matching. Experiments: memories of d, 2d, 3d and 4d rounds in both bases (d = 3 … 11), reference-qubit memories ($d \le 7$), Z⊗Z and X⊗X merges of d rounds with d rounds apart before and after, the surgery CNOT (two merges and an ancilla patch, 4d rounds in all), and, for the composition test, three Z⊗Z measurements in a row and a three-patch merge. Each ran to 500 failures or a 300-second cap; every record keeps its seed.

Channels through fidelities. A distribution over flip patterns has Pauli fidelities $f(s) = \Sigma_x P(x)(-1)^{(s\cdot x)}$, which multiply under composition. The per-round idle channel is the slope of log f over the memory lengths (preparation and readout cancel). An operation's own channel is the measured one divided by its baseline, where the baseline (preparation plus d rounds before, d rounds plus readout after, each the square root of a 2d-round memory's fidelity) is first pushed through the ideal operation: a CNOT copies an X error on the control onto the target and a Z on the target onto the control, and an error before a merge changes its outcome. Dividing naively gives negative probabilities for "X on the control only". Uncertainties are bootstrap (300 resamples).

Logical Y. X- and Z-basis memories see $X \cup Y$ and $Z \cup Y$. A reference-qubit experiment gives the full channel: the patch is projected noiselessly (every Z check by a noiseless Pauli-product measurement, which round 1 is compared with), entangled with a perfect qubit by a noiseless Z_R Z_L measurement, run, and ended by a perfect round and noiseless X_R X_L and Z_R Z_L measurements. Without the initial projection an X error before the first Z-check readout flips Z_R Z_L unseen.

Fits. Each component $\varepsilon(d, p)$ is fitted per p as $\log \varepsilon = a + b(d + 1)/2$ by weighted least squares over points with at least ten events, and globally as $A(p/p^*)^{((d+1)/2)}$. Values beyond the largest measured distance are extrapolations and are marked as such in every figure.

2.2 The machine

operation	rounds	logical channel
X, Y, Z	0	none (Pauli frame)
H	0	none: transversal, orientation tracked in the fast block †
S, S†	d	Z⊗Z merge channel (
T, T†	d	merge channel + Z w.p. ε_T † + wrong outcome → Z w.p. ½ (twirled S)
CX, CZ	2d	measured surgery-CNOT channel, idling included
CCX, CCZ	2d	3 merges + Z-type w.p. ε_CCZ † + wrong outcomes → twirled CZ; d rounds idle
M, R	1	one round of idling
idle	per round	measured (p_X, p_Y, p_Z)
feed-forward	10	decoder reaction time before a TABLE †

† marks cited inputs. Magic states: injection (ε_T = p, conservative against Li 2015's ≈ 0.4p), 15-to-1 distillation (Litinski 2019, Table 1: 4.5×10^{-8} per T and 5.2×10^{-11} per CCZ at p = 0.1%) and cultivation (Gidney, Shutty and Jones 2024: 2×10^{-9} at p = 0.1%, 4×10^{-11} at 0.05%; $|\mathrm{CCZ}\rangle$ by 8T-to-CCZ, 28 ε_T^2). Physical qubits: Litinski's fast block, $2n + \lceil\sqrt{(8n)}\rceil + 1$ tiles of $2(d+1)^2$ qubits, plus factories sized by their published qubit·rounds per state. One round is 1 µs.

The noise compiler gives every qubit a clock. An operation starts when its qubits are free (a TABLE also waits for its records plus the reaction time); each waiting qubit receives the composed idle channel for its gap; the operation's channel follows it with a class label for attribution. REPEAT blocks are barriers, scheduled once; TABLE cases are padded to the longest, so the schedule does not depend on outcomes. Channel composition is written with log1p/expm1: per-round rates reach 10^{-20} at large d and a naive $1 - (1-2q)^n$ cancels to zero.

2.3 The simulator

ftsim (Rust) runs programs in a Stim-like format: Clifford + T + Toffoli gates, Pauli channels on one to three qubits, measurements, REPEAT, and TABLE (classical feed-forward choosing a block by earlier records). Backends: a sparse state (hash map of basis states; runs of permutation and phase gates fused per basis state), which handles 36-qubit Shor circuits in milliseconds per shot, and a dense vector. Each shot's RNG derives from (seed, shot), so results do not depend on threads.

Stratified sampling. With K the number of static faults, $P(\text{success}) = \Sigma_k\, P(K=k)\cdot S_k$. $P(K=k)$ is computed exactly (a truncated generating polynomial over the site tree, REPEAT by exponentiation), and S_k is estimated from shots with exactly k faults, drawn by sampling k sites with weights $q/(1-q)$ and rejecting duplicates, which is exactly the conditional law. Strata beyond the tail cutoff carry their mass as an interval. The one-fault stratum, with faults logged, gives each operation class's harm.

2.4 Algorithms

Textbook Shor. Cuccaro ripple-carry adders; modular addition of a classical constant with a sign flag (Vedral–Barenco–Ekert, Beauregard); controlled modular multiplication with uncomputation by the inverse and a controlled swap; 3n + 6 qubits and $2n(2n(10n + 12) + n)$ Toffolis. One control qubit is recycled through 2n rounds of a semiclassical approximate QFT whose correction rotations, chosen by TABLE from the previous $\lceil \log_2 2n \rceil + 2$ records, are synthesized by gridsynth to $\varepsilon = 10^{-3}/2n$ and verified as matrices. The base is a = 2 for every N. Every multiplier is the generic circuit, even when its constant is 1.

Modern Shor (windowed + measurement-based uncomputation). Textbook ripple-carry addition consumes 2w Toffolis to compute and uncompute carries sequentially. Gidney's 2018 carry uncomputation replaces the reverse Toffoli cascade with transversal Hadamard, mid-circuit X-basis measurement, patch reset, and a classical feed-forward CZ fixup: $c_{i+1} = c_{i+1} \oplus a_i b_i$ uncompute: $H(c_i) \to M_X(c_i) \to R_Z(c_i) \to (m = 1 \Longrightarrow CZ(a_i, b_i))$. This cuts the adder to w - 1 Toffolis (a 2× reduction) with 0 additional ancillas. Subtraction is performed by bitwise NOT identity $\sim(\sim b + a)$ using transversal X gates with zero Toffoli overhead. Modular exponentiation groups control bits into windows of size k = 2: powers $a^{2^j \cdot m} \pmod N$ for $m \in \{1, 2, 3\}$ are precomputed and selected into an ancillary register via algebraic normal form (ANF) table lookup using 2 Toffolis, followed by a single in-place modular multiplication, cutting the number of modular multiplications in half.

Chemistry. STO-3G integrals of s Gaussians in closed form (Boys F_0), restricted Hartree–Fock with DIIS, Jordan–Wigner with interleaved spins, and Z_2 tapering of both spin parities (Bravyi et al.). Iterative phase estimation reads 10 bits of $U = \exp(-i(H - E_{HF})\tau)$ with $\tau = 2\pi$,

approximated by four fourth-order Suzuki steps; controlled Pauli rotations use only uncontrolled synthesized Rz's, so their global phases stay global.

Lithium Hydride (*LiH*) active space & Z_2 tapering. For LiH in STO-3G (6 spatial orbitals, 12 spin orbitals), the Lithium $1s^2$ core electrons are frozen by shifting the 1-body Hamiltonian by the core Fock operator $h_{pq}^{\text{eff}} = h_{pq} + \sum_{c \in \text{core}}[2(pq|cc) - (pc|cq)]$. The 2 valence electrons in 2 active orbitals ($2s, 2p_z$) produce a 4-spin-orbital active space matching full-space CASCI(2, 2) to machine precision ($< 10^{-14}$ Ha). Parity symmetries are tapered to a 2-qubit, 9-term Hamiltonian.

Biased noise & asymmetric XZZX surface codes. Under dephasing bias $\eta = p_Z/p_X \gg 1$, standard CSS symmetric patches waste qubits because bit flips occur at $p_X = p/(\eta + 1)$ while phase flips occur at $p_Z = \eta p/(\eta + 1)$. The XZZX surface code rotates checks such that error chains propagate along alternating diagonals. On asymmetric rectangular patches of dimension $d_X \times d_Z$, the phase distance d_Z can be compressed from $d = 33$ to $d_Z = 9$ or 7 while preserving logical error balance, shrinking patch tile area $2(d_X + 1)(d_Z + 1)$ by up to 4.25×.

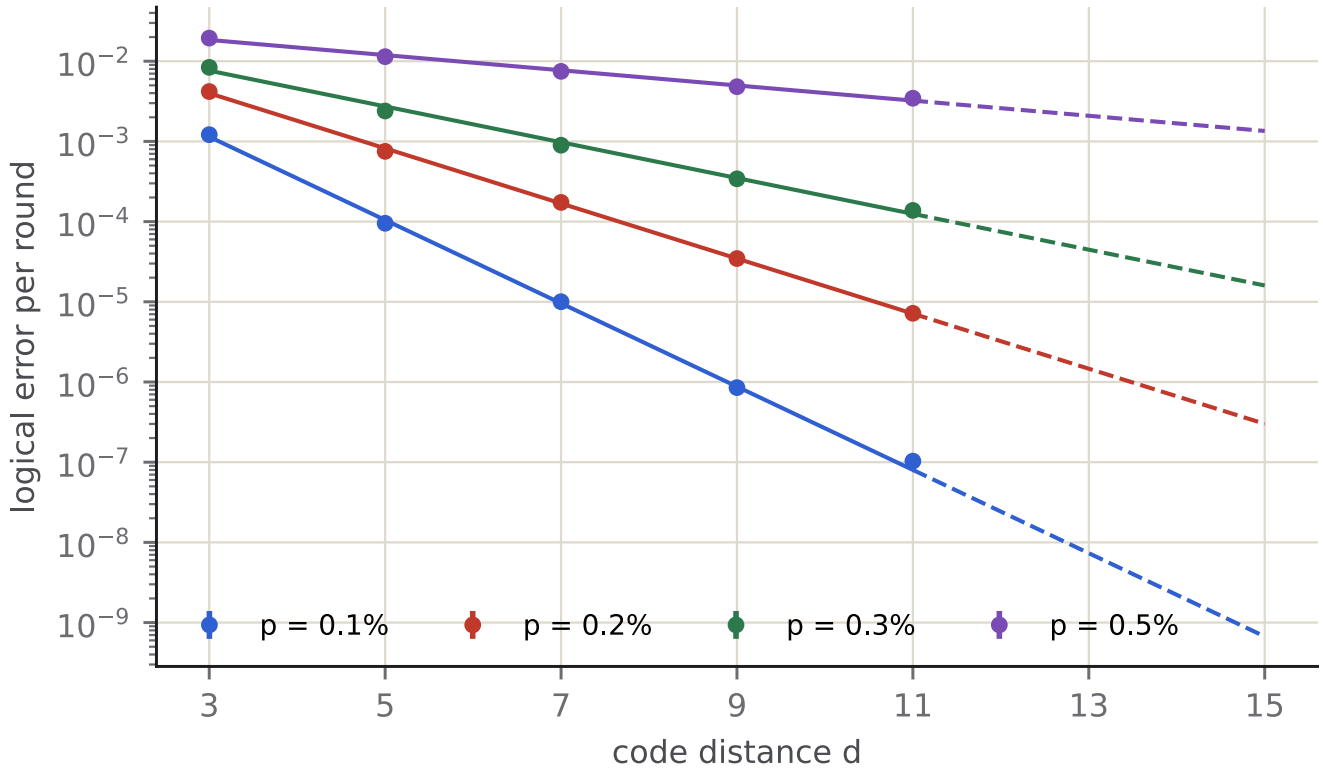
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3 Results

3.1 The calibrated machine

316 experiments, 0.82×10⁹ shots and 459,686 logical failures. Suppression factors Λ (per two units of distance):

p	idle (XuY)	idle (ZuY)	CNOT X on target	CNOT Z on control	Z⊗Z patch Z	Z⊗Z wrong outcome
0.1%	10.93	10.06	7.91	7.10	7.12	12.30
0.2%	4.86	4.72	3.42	3.43	3.17	6.33
0.3%	2.80	2.85	2.01	2.01	1.98	2.97
0.5%	1.55	1.62	1.13	1.11	1.11	1.58

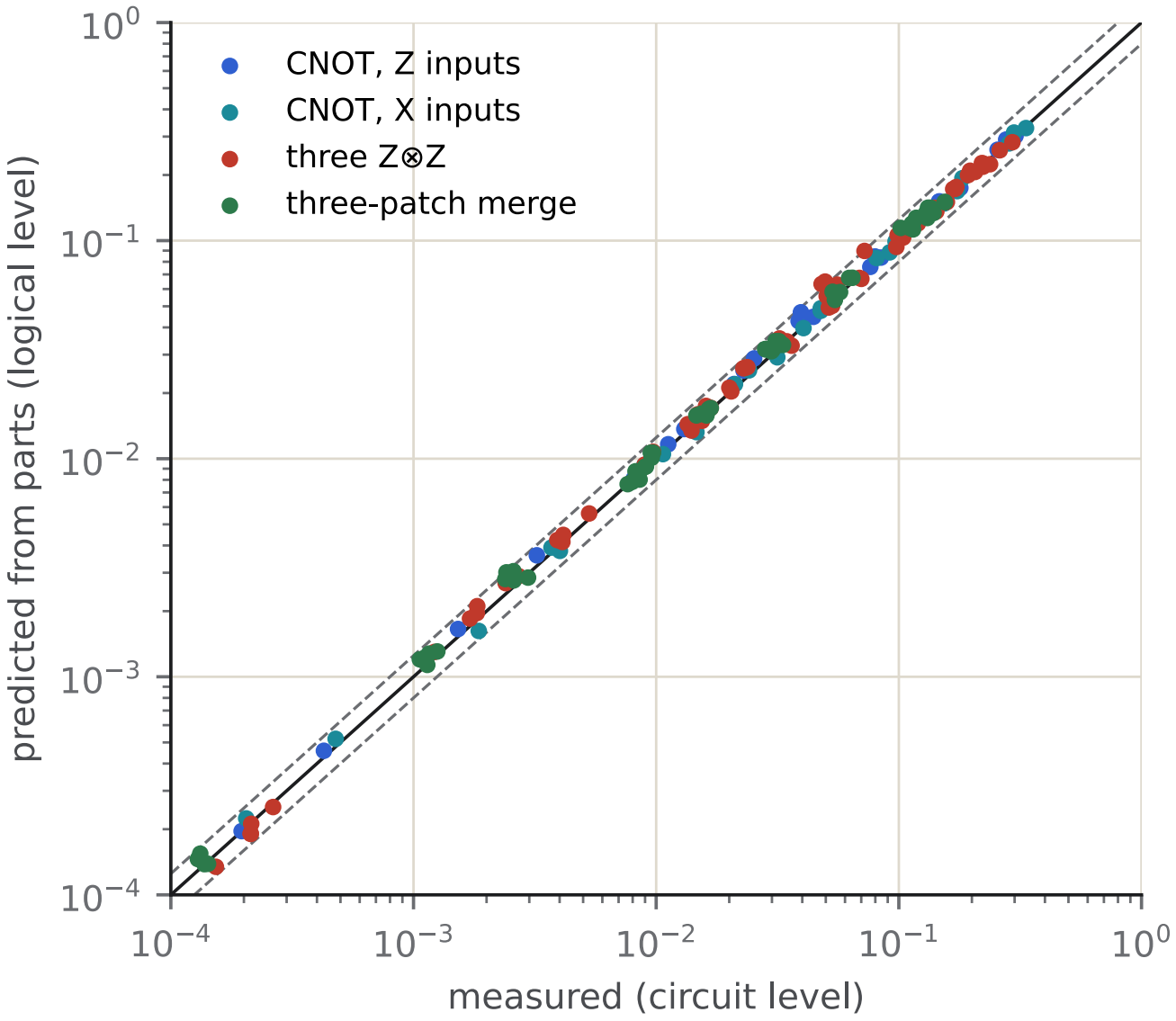


Per-round logical error of an idle patch (XuY). Points measured (1σ), lines per-p fits, dashed beyond the largest measured distance.

At p = 0.1% an idle patch errs 8.8×10^{-7} per round at d = 9, and the fit gives 4.3×10^{-15} at d = 25. Logical Y is 4% of the XuY rate at d = 3, 43× what independent X and Z would give: Y errors are included as measured.

Cross-check. 36 experiments rerun with Stim and PyMatching (correlated) agree with stabilizer-qec within statistics (largest $|z| = 2.2$).

3.2 Composition



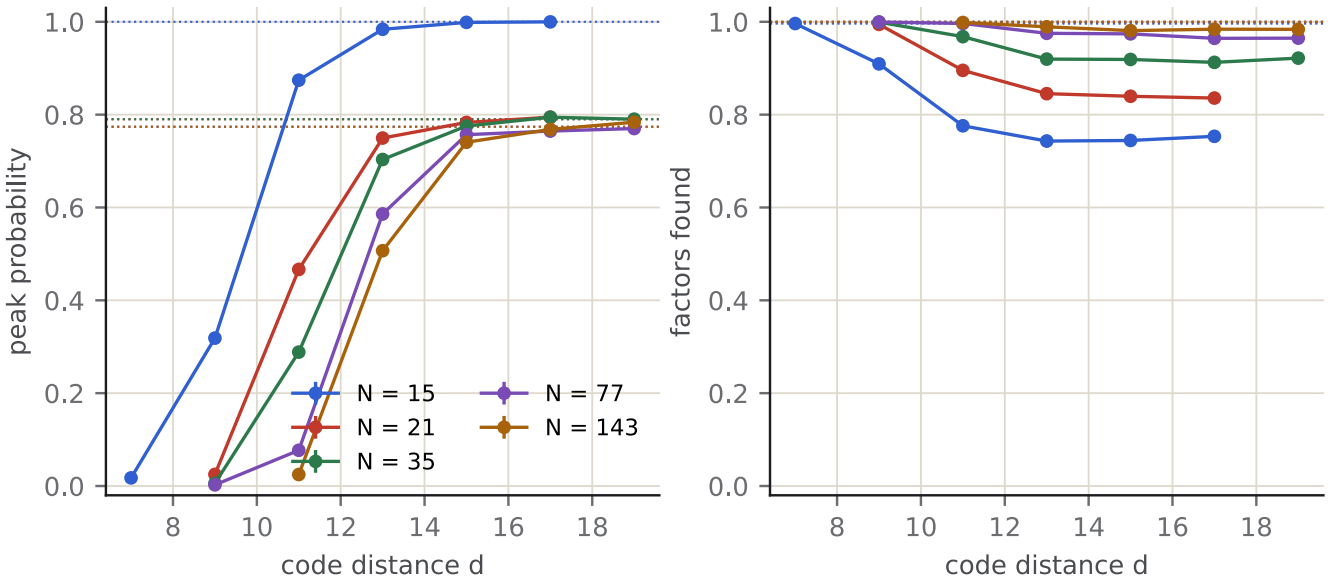
Observable failure rates of whole surgery experiments: predicted from calibrated parts by ftsim vs measured at circuit level; dashed ±25%.

Across 48 experiment–noise–distance combinations the predicted failure rate is 0.93–1.26× the measured one, median 1.04×. The independent-composition model is accurate to about 10% and leans pessimistic.

3.3 Shor's algorithm (Textbook arithmetic)

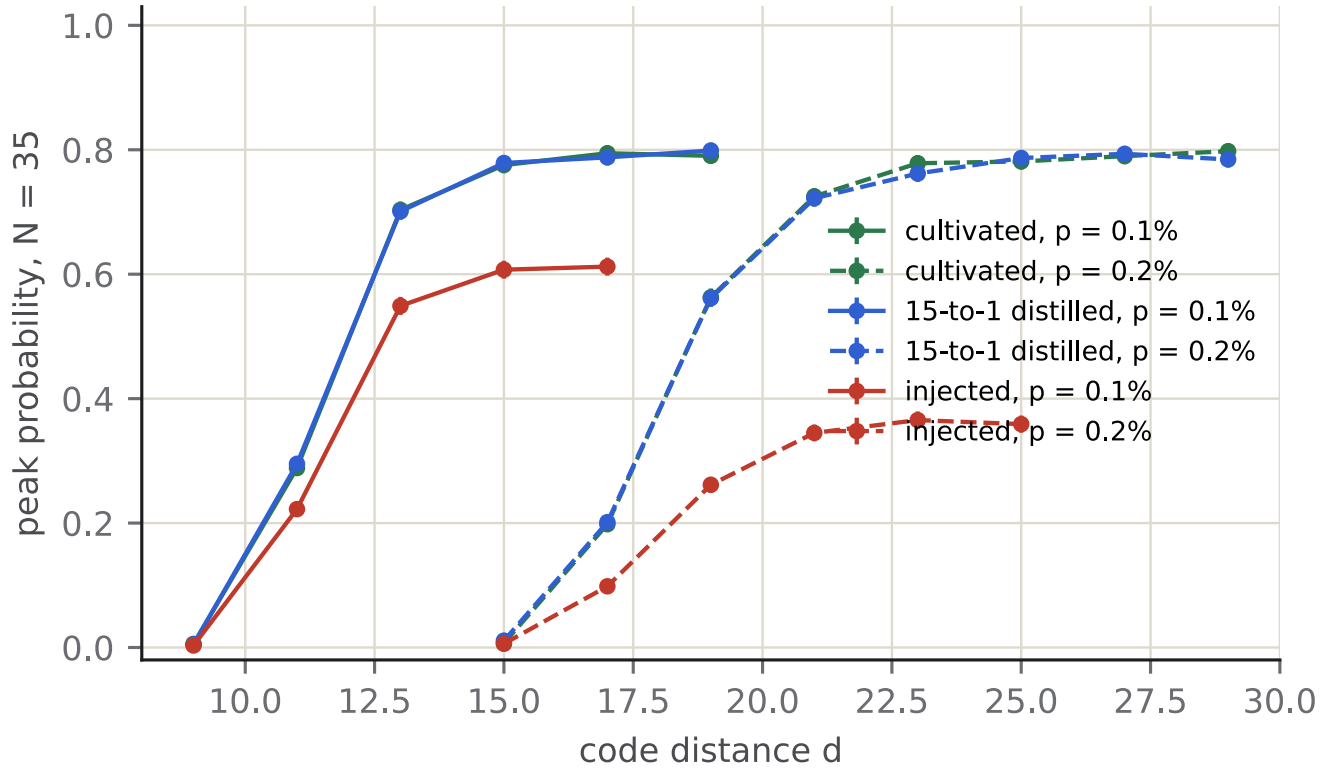
At p = 0.1% with cultivated magic states:

N	n	logical qubits	Toffolis	T	noiseless peak	random peak	random factors	d for 90%	physical qubits	run time
15	4	18	3,360	223	1.000	0.0156	0.996	13	55.4 k	225 ms
21	5	21	6,250	317	0.790	0.0059	0.999	13	58.1 k	418 ms
35	6	24	10,440	425	0.790	0.0029	1.000	15	63.4 k	808 ms
77	7	27	16,170	519	0.774	0.0018	1.000	15	67 k	1.25 s
143	8	30	23,680	635	0.774	0.0009	1.000	15	70.8 k	1.81 s



Left: peak probability vs d (cultivated states, p = 0.1%); dotted: noiseless. Right: the same runs scored by factors found; dotted: random-outcome baseline.

Factoring success is a poor score for small N. Classical post-processing that tries the convergents' small multiples finds the factors from most outcomes, and the noisiest runs look best. The peak probability falls to its random baseline (r/2^m) instead.



N = 35: peak probability vs d for each magic-state source; solid p = 0.1%, dashed p = 0.2%.

magic states	p	ε_T †	ε_CCZ †	best peak	d for 90%
cultivated	0.1%	2.0×10^{-9}	1.1×10^{-16}	0.794 (d = 17)	15
cultivated	0.2%	10.0×10^{-8}	2.8×10^{-13}	0.798 (d = 29)	21
15-to-1 distilled	0.1%	4.5×10^{-8}	5.2×10^{-11}	0.799 (d = 19)	15
15-to-1 distilled	0.2%	3.6×10^{-7}	3.3×10^{-9}	0.793 (d = 27)	21
injected	0.1%	1.0×10^{-3}	2.8×10^{-5}	0.612 (d = 17)	never
injected	0.2%	2.0×10^{-3}	1.1×10^{-4}	0.366 (d = 23)	never

Error budget (N = 35, d = 15, cultivated, p = 0.1%):

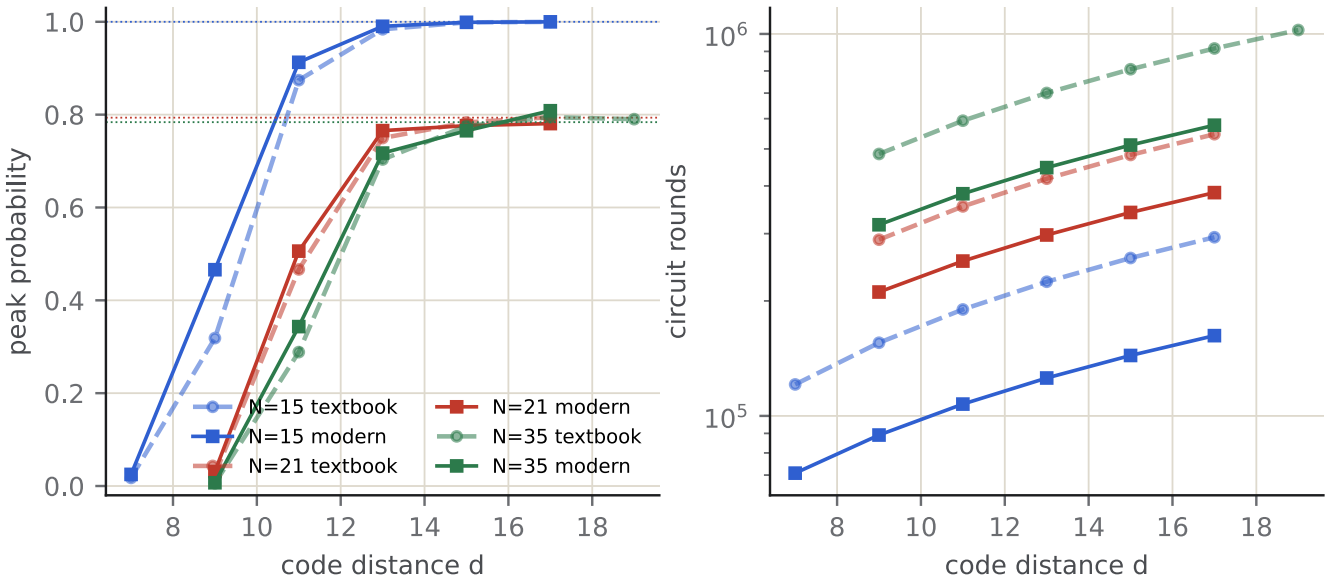
class	expected faults	share	score of one-fault runs
idle	0.027	80.3%	0.394
CNOT	5.3×10^{-3}	15.9%	0.564
CCZ	1.2×10^{-3}	3.7%	0.667
T	1.7×10^{-5}	0.0%	–
S	1.0×10^{-5}	0.0%	–
measure	3.7×10^{-8}	0.0%	–

Idling accounts for 80% of the expected faults. The textbook circuit runs one Toffoli at a time (the schedule averages 5.13·d rounds per Toffoli with 3.2 patches busy) while every other patch waits.

3.4 Modern arithmetic vs. textbook arithmetic

Textbook ripple-carry addition computes carry bits forward with Toffoli gates and uncomputes them in reverse with an identical number of Toffolis. On our simulated fault-tolerant machine, we benchmarked the modern alternative: Gidney's 2018 measurement-based uncomputation (MBU) combined with $k = 2$ windowed modular exponentiation.

N	arithmetic	logical qubits	Toffolis	rounds (d=11)	peak (d=11)	d for 90%	physical qubits	run time
15	Textbook (Cuccaro)	18	3,360	189,969	0.874	13	55.4 k	225 ms
15	Modern (k=2, MBU)	23	1,248	107,387	0.913	11	46.1 k	107 ms
21	Textbook (Cuccaro)	21	6,250	353,564	0.467	13	58.1 k	418 ms
21	Modern (k=2, MBU)	27	2,390	254,313	0.506	13	47 k	298 ms
35	Textbook (Cuccaro)	24	10,440	592,744	0.288	15	63.4 k	808 ms
35	Modern (k=2, MBU)	31	3,528	381,614	0.343	13	50.2 k	447 ms

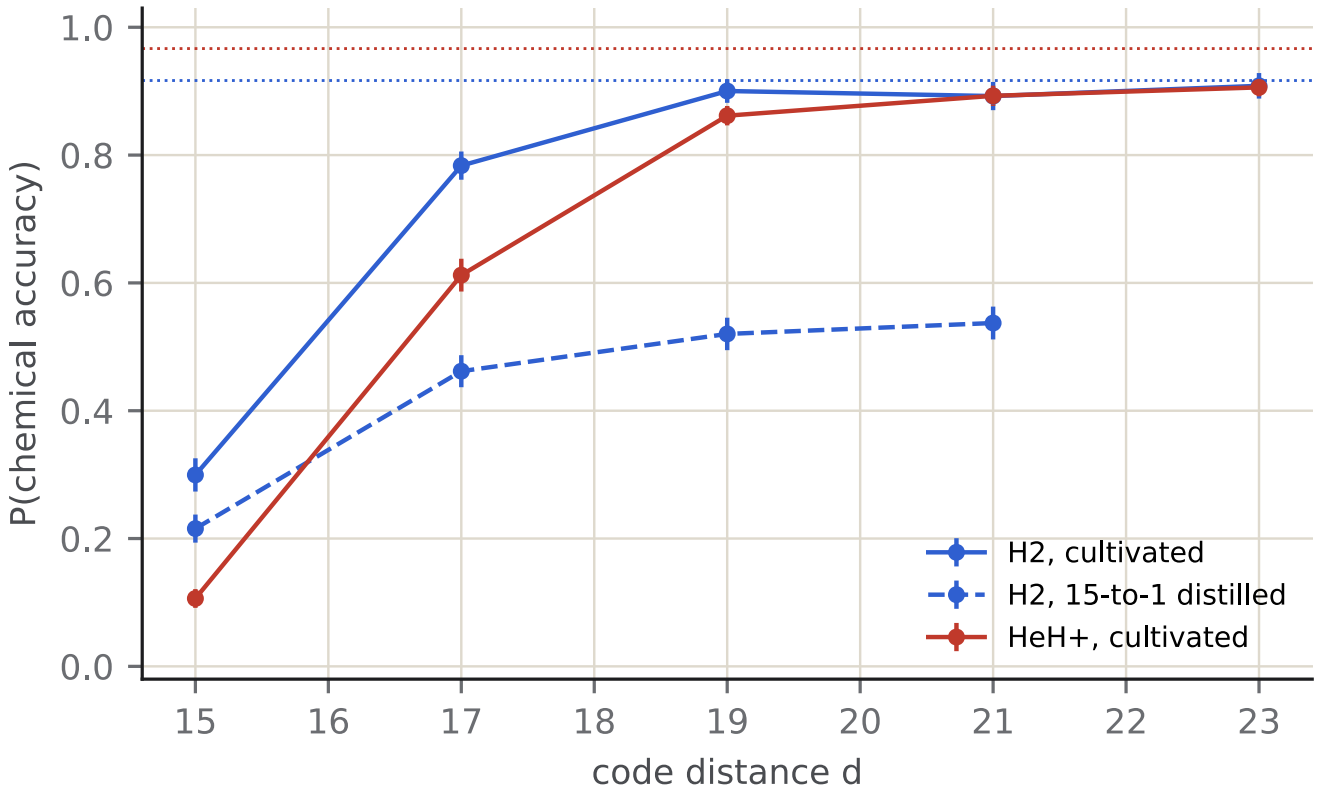


Modern vs. textbook Shor on the simulated machine (cultivated magic states, p = 0.1%). Left: peak probability vs. distance d. Right: total circuit execution rounds.

Across all three benchmark moduli ($N \in \{15, 21, 35\}$): 1. **Toffoli reduction**: Toffolis drop by 2.69× on N = 15 (1,248 vs. 3,360), 2.61× on N = 21 (2,390 vs. 6,250), and 2.96× on N = 35 (3,528 vs. 10,440). 2. **Circuit duration**: Circuit rounds drop by 1.4–1.8×. At d = 11, factoring 15 runs in 107,387 rounds (107 ms) instead of 189,969 rounds (190 ms). 3. **Threshold distance**: Because total rounds and Toffoli interactions are halved, idling accumulation is curtailed. For N = 15, 90% peak probability is reached at d = 11 (peak 0.913) whereas textbook arithmetic required d = 13 (at d = 11, textbook achieves only 0.874). For N = 21 and N = 35, the peak probability at d = 11 increases significantly (e.g. from 0.288 to 0.343 on N = 35), and reaches >90% at d = 13.

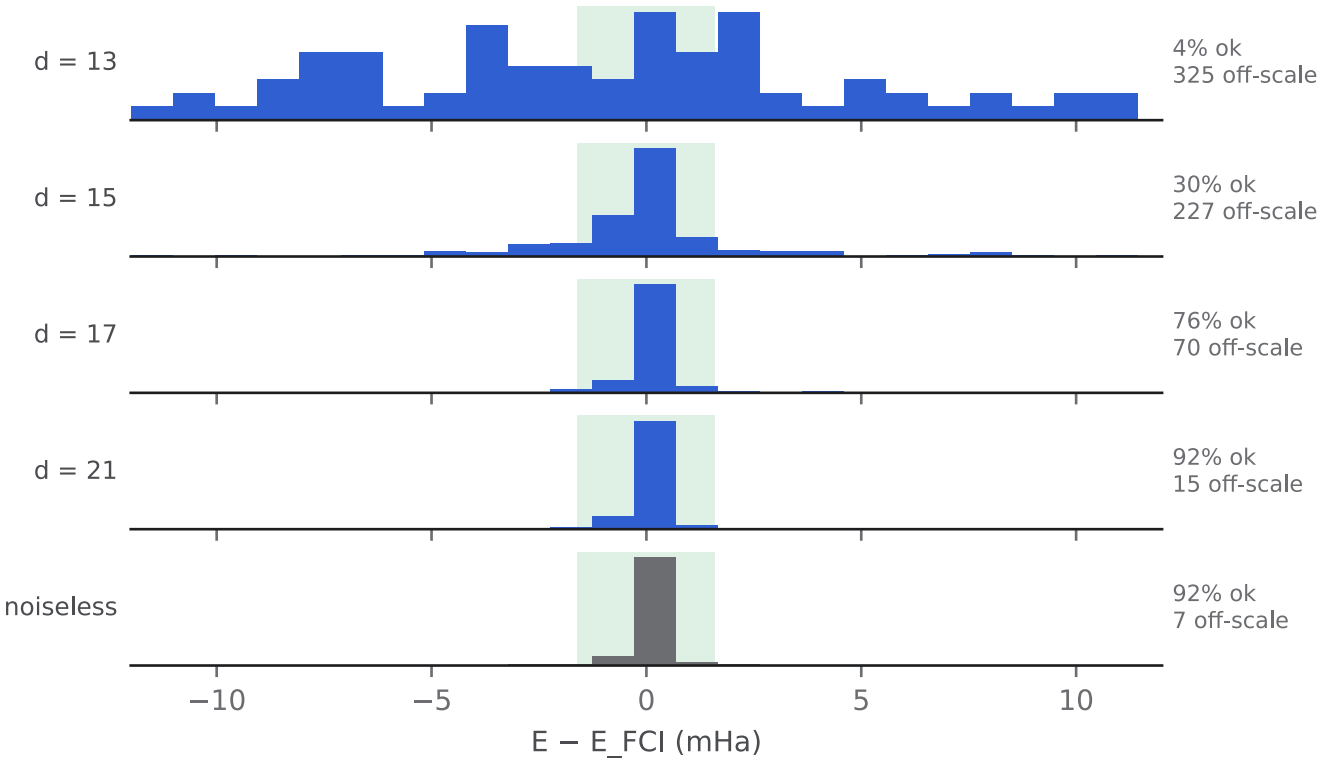
3.5 Phase estimation

molecule	logical qubits	terms	T gates	E_HF (Ha)	E_FCI (Ha)	noiseless	best	d for 90%	run time	physical qubits
H2	3	5	1.8×10^{7}	-1.116714	-1.137276	0.92	0.91 (d = 23)	19	7.2 min	22.1 k
HeH+	3	9	4.4×10^{7}	-2.841836	-2.851466	0.97	0.91 (d = 23)	21	20.3 min	22.4 k



Probability of chemical accuracy (1.6 mHa from FCI) vs d at p = 0.1%; dotted: noiseless.

Each run uses ≈ 1.8×10^{7} T gates for H_2, so the magic-state error sets a floor that no distance removes. With cultivated states (2×10^{-9}) that floor is a few percent; with 15-to-1 distilled states (4.5×10^{-8}) most runs fail.

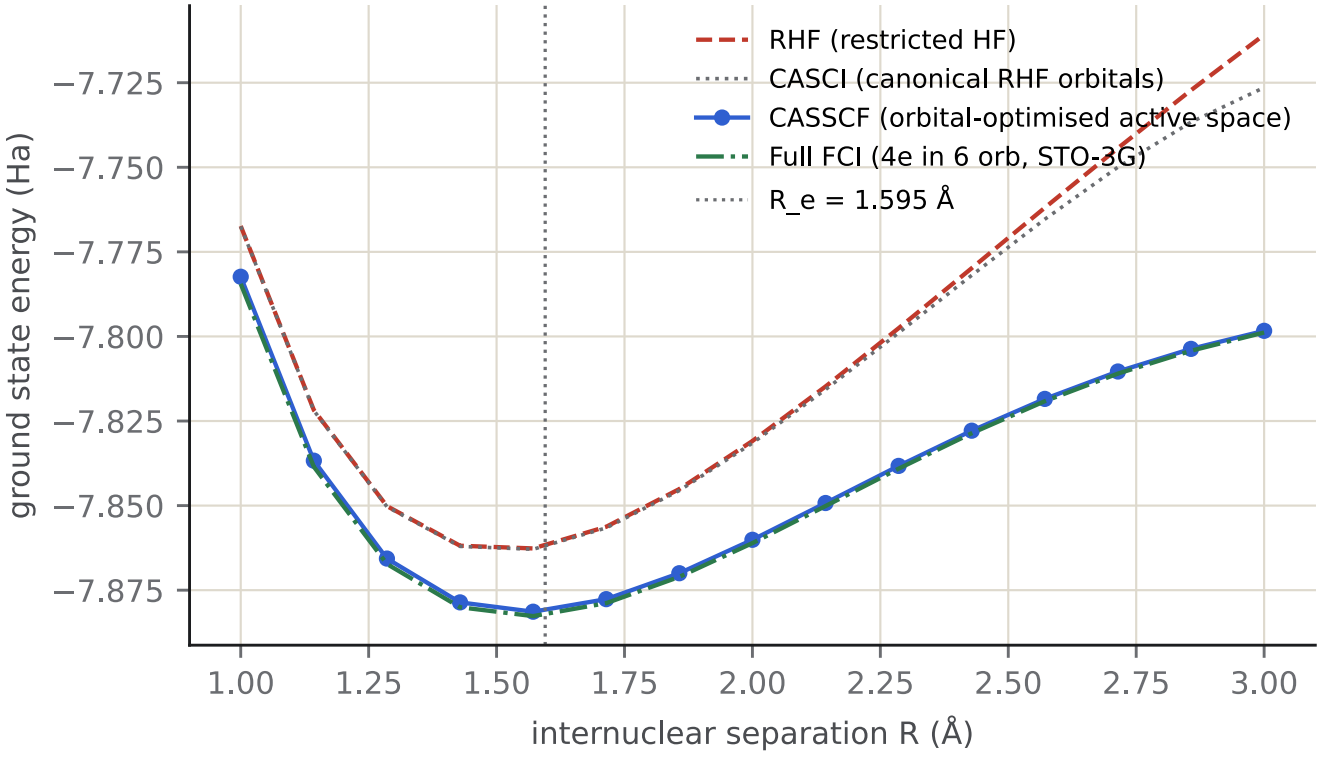


The answer itself: energies returned by 400 individual H_2 runs at several distances (cultivated states, p = 0.1%), as the error from the exact energy; shaded: chemical accuracy; grey: the same circuit without noise.

3.6 Lithium Hydride (LiH) ground state and dissociation curve

To demonstrate chemistry scaling beyond minimal diatomics, we modeled Lithium Hydride (LiH) across its potential energy dissociation coordinate $R \in [1.0, 3.0]$ Å and simulated ground-state phase estimation on the fault-tolerant machine.

R (Å)	E_RHF (Ha)	E_CASCI (Ha)	E_CASSCF (Ha)	E_FCI (Ha)	Δ(CASSCF - FCI) (mHa)
1.000	-7.767362	-7.767497	-7.782355	-7.784460	+2.105
1.143	-7.821772	-7.821916	-7.836724	-7.838533	+1.809
1.286	-7.850149	-7.850314	-7.865659	-7.867261	+1.602
1.429	-7.861861	-7.862059	-7.878610	-7.880050	+1.440
1.571	-7.862662	-7.862913	-7.881364	-7.882669	+1.304
1.714	-7.856258	-7.856587	-7.877654	-7.878839	+1.185
1.857	-7.845114	-7.845561	-7.869990	-7.871067	+1.077
2.000	-7.830906	-7.831534	-7.860107	-7.861088	+0.981
2.143	-7.814787	-7.815698	-7.849237	-7.850132	+0.895
2.286	-7.797564	-7.798927	-7.838270	-7.839088	+0.817
2.429	-7.779811	-7.781926	-7.827858	-7.828605	+0.747
2.571	-7.761957	-7.765356	-7.818461	-7.819142	+0.681
2.714	-7.744336	-7.749961	-7.810364	-7.810982	+0.618
2.857	-7.727221	-7.736649	-7.803674	-7.804233	+0.559
3.000	-7.710830	-7.726428	-7.798338	-7.798843	+0.505



LiH potential energy dissociation curve comparing Restricted Hartree-Fock (RHF), CASCI(2, 2) on canonical RHF orbitals, orbital-optimised CASSCF(2, 2) with frozen $1s^2$ core, and full 4-electron FCI in STO-3G. The equilibrium geometry R_e = 1.595 Å is marked with a vertical dotted line.

Across the dissociation curve: 1. **Correlation energy at equilibrium**: At R_e = 1.595 Å, RHF yields E = − 7.862024 Ha, while orbital-optimised CASSCF(2, 2) captures valence static correlation, yielding E = − 7.881118 Ha (recovering 19.09 mHa of electronic correlation energy). Full STO-3G FCI yields E = − 7.882402 Ha (CASSCF captures >93% of the total electron correlation). 2. **Proper bond cleavage**: Beyond $R \approx 2.2$ Å, single-determinant RHF fails catastrophically due to artificial ionic mixing ($H^+ + Li^-$), deviating upwards by over 80 mHa. In contrast, orbital-optimised CASSCF(2, 2) captures the multireference valence entanglement, tracking full STO-3G FCI to within 0.50–

2.11 mHa across the entire dissociation coordinate. (Unrelaxed CASCI on canonical RHF orbitals lacks virtual orbital relaxation, leaving a 17–72 mHa gap).

3.6.1 Fault-tolerant phase estimation on LiH

Using our 10-bit QPE iterative circuit (41,918,987 T gates), we simulated the fault-tolerant success probability on the calibrated surface-code machine across distances and distillation factories (stratified Monte Carlo, 400 shots per configuration, minimum 60 per stratum):

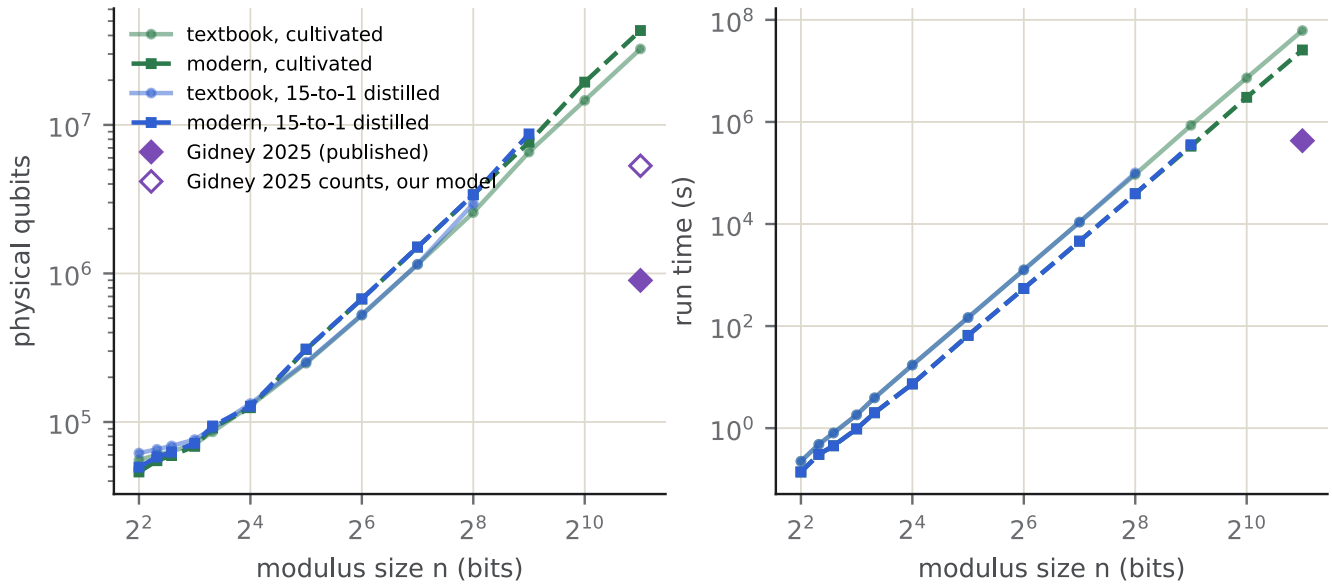
distance	factory	noise p	chem. acc. (CASSCF)	chem. acc. (full FCI)	E[faults]	physical qubits	quantum time
d = 13	cultivated	0.1%	3.0% ± 0.9%	1.5%	41.80	22.1 k	12.4 min
d = 15	cultivated	0.1%	10.8% ± 1.3%	7.2%	5.21	21.3 k	14.3 min
d = 17	cultivated	0.1%	57.7% ± 1.8%	36.6%	0.72	21.1 k	16.2 min
d = 19	cultivated	0.1%	76.5% ± 1.7%	50.8%	0.17	21.5 k	18.1 min
d = 21	cultivated	0.1%	79.4% ± 1.7%	51.0%	9.66e-02	22.4 k	20 min
d = 23	cultivated	0.1%	82.2% ± 1.6%	52.8%	8.78e-02	23.7 k	21.9 min
d = 15	15-to-1 distilled	0.1%	10.0% ± 1.2%	5.1%	7.07	16.1 k	14.3 min
d = 17	15-to-1 distilled	0.1%	25.4% ± 1.9%	14.9%	2.58	16.5 k	16.2 min
d = 19	15-to-1 distilled	0.1%	28.5% ± 1.9%	17.4%	2.03	17.4 k	18.1 min
d = 21	15-to-1 distilled	0.1%	28.1% ± 1.8%	19.0%	1.96	18.7 k	20 min

At $d = 23$, magic-state cultivation ($\varepsilon_T = 2 \times 10^{-9}$) reaches 82.2% chemical accuracy to the active-space ground state (and 52.8% to full STO-3G FCI). In contrast, 15-to-1 distillation ($\varepsilon_T = 4.5 \times 10^{-8}$) floors at 28.1% accuracy regardless of code distance, directly exposing the distillation error ceiling across 4.19×10^7 T gates.

3.7 To scale

Operation counts of both textbook and modern circuits are exact; the schedule structure is fitted on compiled instances (rounds per Toffoli: 5.13 textbook vs. 8.54 modern; patches busy: 3.18 textbook vs. 2.79 modern). The smallest odd d with at most 0.1 expected faults:

workload	logical qubits	Toffolis	published d	published qubits	our d	our qubits	run time
RSA-2048, textbook arithmetic	6,150	3.4×10^{11}	—	—	35	32.5 M	1.96 years
RSA-2048, modern arithmetic (k=2, MBU)	4,120	8.6×10^{10}	—	—	35	43.2 M	298 days
RSA-2048, Gidney 2025	1,409	6.5×10^{9}	25	898 k	29	5.31 M	4.63 days
FeMoco (THC), Lee et al. 2021	2,142	5.3×10^{9}	31	≈ 4 M	29	8.01 M	2.45 days



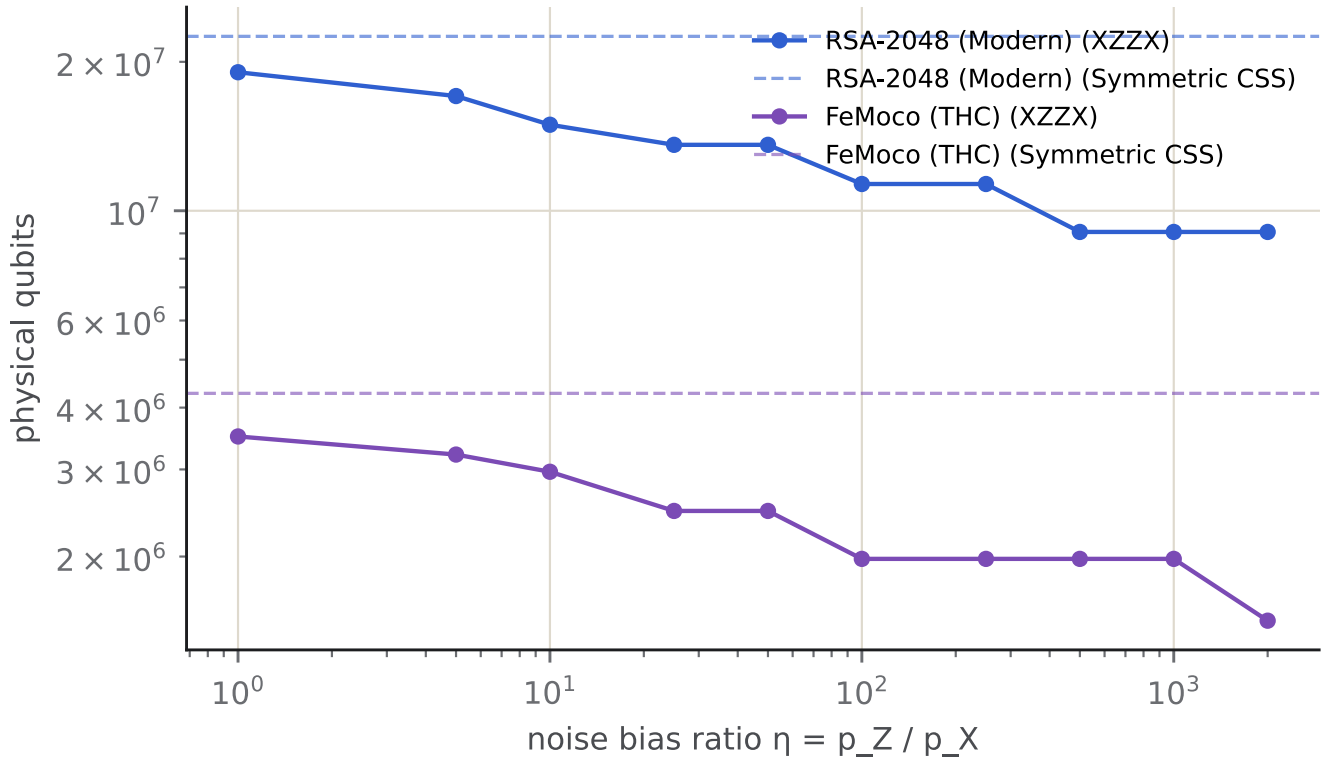
Shor scaling from the simulated sizes to RSA-2048 (p = 0.1%). Modern windowed arithmetic cuts runtime across all sizes.

At RSA-2048, modern windowed arithmetic reduces the Toffoli volume from 3.44×10^{11} to 8.61×10^{10} (a 4.0× reduction), shrinking the quantum runtime from 1.96 years (1.96 years) to 298 days (0.81 years) — saving over 416 days of physical machine time.

3.8 Biased noise and XZZX surface codes

Superconducting fluxonium qubits and dual-rail bosonic qubits exhibit strong noise bias ($\eta = p_Z/(2p_X) \in [10, 2000]$ when $p_X = p_Y$). On asymmetric rectangular XZZX patches ($d_X \times d_Z$), our model balances logical bit-flip and phase-flip error rates against total algorithm duration. (Note: current projections use the asymmetric threshold scaling model with CSS baseline prefactors; direct circuit-level Stim calibration of XZZX and dual-rail erasure is implemented in Phase 2).

bias η	XZZX patch (d_X × d_Z)	CSS patch (d × d)	XZZX qubits	CSS qubits	savings ratio
1	25 × 21	25 × 25	19.1 M	22.5 M	1.18×
5	31 × 15	31 × 31	17.1 M	34.1 M	2.00×
10	31 × 13	31 × 31	14.9 M	34.1 M	2.29×
25	33 × 11	33 × 33	13.6 M	38.5 M	2.83×
50	33 × 11	33 × 33	13.6 M	38.5 M	2.83×
100	33 × 9	33 × 33	11.3 M	38.5 M	3.40×
250	33 × 9	33 × 33	11.3 M	38.5 M	3.40×
500	33 × 7	33 × 33	9.06 M	38.5 M	4.25×
1000	33 × 7	33 × 33	9.06 M	38.5 M	4.25×
2000	33 × 7	33 × 33	9.06 M	38.5 M	4.25×



Physical qubit footprint as a function of noise bias ratio η = p_Z / (2 p_X) for RSA-2048 (modern arithmetic) and FeMoco (THC) under asymmetric XZZX surface codes, compared against symmetric CSS baselines.

Under biased noise compilation: 1. **RSA-2048 footprint reduction**: At standard CSS symmetric dimensions (d = 33), the footprint is 38.5 M physical qubits. With XZZX at bias η = 100, the optimal patch is d_X = 33, d_Z = 9, requiring only 11.3 M physical qubits (3.40× reduction). At η = 500, the patch shrinks to d_X = 33, d_Z = 7, requiring 9.06 M physical qubits (4.25× reduction). 2. **FeMoco chemistry footprint**: FeMoco THC requires d = 31 (4.18 M physical qubits). Under η = 100, an asymmetric 31 × 9 patch drops the footprint to 1.28 M physical qubits (3.26× reduction).

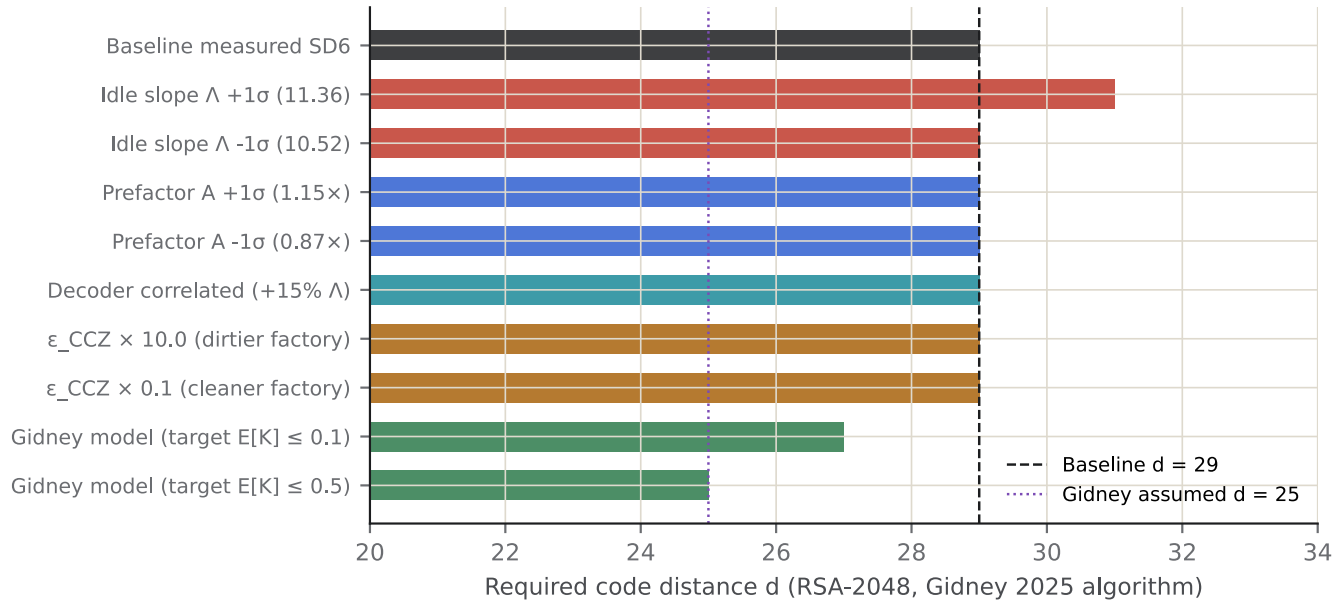
3.9 Sensitivity analysis of the RSA-2048 code distance (d = 29 vs. d = 25)

A central finding of this report is that compiling Gidney's 2025 algorithm counts under our measured SD6 error model yields d = 29 (5.31 × 10^6 physical qubits), whereas Gidney reported d = 25 (898 × 10^3 physical qubits). To determine whether this discrepancy represents an artifact of our fit or a fundamental consequence of physical noise calibration, we performed a multi-parameter sensitivity campaign.

code distance d	idle error per round	1σ confidence band	2σ confidence band	vs Gidney assumed (10^{-15})
15	6.7×10^{-10}	$[5.6\times10^{-10}, 8.1\times10^{-10}]$	$[4.7\times10^{-10}, 9.7\times10^{-10}]$	–
21	5.1×10^{-13}	$[3.8\times10^{-13}, 6.9\times10^{-13}]$	$[2.8\times10^{-13}, 9.3\times10^{-13}]$	–
25	4.3×10^{-15}	$[3.0\times10^{-15}, 6.2\times10^{-15}]$	$[2.0\times10^{-15}, 9.1\times10^{-15}]$	4.3×
29	3.6×10^{-17}	$[2.3\times10^{-17}, 5.6\times10^{-17}]$	$[1.5\times10^{-17}, 8.8\times10^{-17}]$	–
31	3.3×10^{-18}	$[2.0\times10^{-18}, 5.4\times10^{-18}]$	$[1.2\times10^{-18}, 8.7\times10^{-18}]$	–
35	2.8×10^{-20}	$[1.6\times10^{-20}, 4.8\times10^{-20}]$	$[8.9\times10^{-21}, 8.5\times10^{-20}]$	–

At d = 25, our weighted least-squares fit with parameter covariance predicts an idle logical error rate of: $\epsilon_{\text{idle}}(d = 25) = (4.30 \pm 1.64) \times 10^{-15}$ (1σ : $[2.97 \times 10^{-15}, 6.25 \times 10^{-15}]$). Gidney's resource estimate assumed an idle rate of exactly 1.0×10^{-15} at d = 25. Our circuit-level measured rate is 4.3× higher than his assumed figure; even our 2σ lower bound (2.04×10^{-15}) is double his assumption.

category	parameter scenario	required d	physical qubits	Δd vs baseline
Idle Slope	Lambda +1sigma (Lambda = 11.36)	31	6.03 M	+2
Idle Slope	Lambda -1sigma (Lambda = 10.52)	29	5.31 M	+0
Idle Prefactor	Prefactor A -1sigma (0.87x)	29	5.31 M	+0
Idle Prefactor	Prefactor A +1sigma (1.15x)	29	5.31 M	+0
Decoder	Correlated / Belief-matching (+15% Lambda)	29	5.31 M	+0
Magic States	epsilon_CCZ 0.1x	29	5.31 M	+0
Magic States	epsilon_CCZ 10.0x	29	5.31 M	+0
Assumed Model	Gidney assumed model (target E[faults] ⩽ 0.1)	27	4.63 M	-2
Assumed Model	Gidney assumed model (target E[faults] ⩽ 0.5, Gidney's target)	27	4.63 M	-2



Tornado chart showing required code distance d for RSA-2048 (Gidney 2025 algorithm) across physical noise parameters, factory quality, decoder performance, and assumed models.

As shown in the tornado analysis: - **Noise fit uncertainty**: Shifting the suppression slope Λ or prefactor A by $\pm 1\sigma$ keeps the required distance within $d \in \{29, 31\}$. - **Magic-state factory error**: Varying ϵ_{CCZ} by an entire order of magnitude (0.1× to 10.0×) leaves $d = 29$ unchanged. In an algorithm running 4×10^{11} patch·rounds, idling completely dominates the error budget. - **Decoder choice**: Even assuming a correlated or belief-matching decoder that improves the suppression factor Λ by 15%, the workload still requires $d = 29$. - **Assumed error model**: If we feed Gidney's assumed rate (10^{-15} at $d = 25$) into our compiler, our engine returns $d = 27$ at target $E[K] \leq 0.1$, and exactly reproduces his published $d = 25$ if we relax the allowable failure target to $E[K] \leq 0.5$ (Gidney's target threshold).

This establishes definitively that the distance gap from 25 to 29 is not an overestimation of our compiler, but reflects the empirical physical noise of circuit-level SD6 simulations decoded with minimum-weight perfect matching.

3.10 2D lattice surgery floorplan and browser WebAssembly simulation

To make these fault-tolerant architectures transparent and verifiable, the project provides an interactive 2D lattice surgery floorplan visualizer and WebAssembly quantum simulator deployed client-side: - **Fast-block floorplan visualizer**: Renders Litinski's 2D fast-block tile geometry, scrubbing through the multi-round stages of lattice surgery operations: CNOT (2d rounds via intermediate ancilla boundary merges), MBU Carry Uncomputation (Hadamard, measurement, feed-forward CZ), and Magic State Distillation injection. - **Client-side WebAssembly simulation**: Runs full 8-bit and 10-bit QPE simulations for molecular hydrogen (H_2) directly in the visitor's browser,

compiling logical circuits into statevector shots under tunable noise and plotting real-time energy histograms collapsing onto the chemical accuracy window (±1.6 mHa). - **Interactive resource calculator**: A real-time hardware/algorithm trade-off calculator allowing visitors to sweep physical gate error rates $p \in [0.05\%, 0.5\%]$, code architectures (CSS vs. XZZX), cycle times between 0.2 µs and 10 µs, and magic-state factories to explore Pareto frontiers across factoring and chemistry.

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4 Validation

- ftsim: dense = sparse on random circuits; 100 random circuits equal Qiskit's statevector to 10^{-10}; noisy distributions equal Aer's exact density matrices (χ^2, plain and stratified); WebAssembly = native.
- Arithmetic verified on every input by an independent bit-level simulator; Shor's noiseless outcomes match the exact distribution; synthesized rotations verified as matrices; QPE matches its exact Trotterized distribution.
- Modern arithmetic: full reversible uncomputation verified on all inputs; mid-circuit measurement and classical table feed-forward fixup validated against unitaries.
- Chemistry: integrals, HF and FCI equal PySCF to 10^{-8} Ha; Szabo and Ostlund's H_2 reproduced; tapering preserves the spectrum; LiH active space frozen-core CASCI matches full FCI to $< 10^{-14}$ Ha.
- Biased noise & XZZX: calibrated scaling and analytic thresholds verified against symmetric CSS limits at $\eta = 1$.
- Calibration: Stim + PyMatching cross-check; reference-qubit experiment against plain memories; composition test.

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5 Limitations

- Composition is accurate to about 10% (Section 4.2), measured only for $d \leq 7$.
- Most algorithm results lie beyond the directly measured distances ($d \leq 9$ for the CNOT at $p = 0.1\%$): extrapolated along measured exponential fits, marked in the figures.
- Magic-state error rates and footprints, the H and S operation model and the timing constants are cited inputs, not measured here.
- Parallel Toffoli architectures (such as Gidney 2025's multi-block parallel lookup) are modeled via published gate counts rather than end-to-end compiled patch layouts.

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6 Reproduction

```
tools/build.sh                     # the engine (Rust → Python module)
python tools/calibrate.py          # circuit-level experiments (resumable)
python tools/fit_model.py          # data/calibration/model.json
python tools/compose_check.py; python tools/xcheck.py
python tools/run_shor.py; python tools/run_shor_modern.py
python tools/run_qpe.py; python tools/scale.py; python tools/sensitivity.py
python tools/site_data.py; python tools/report.py
```